Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The vatue of the integral, $\int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}} dx$ is

(A) 1

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$

(D) 2

Q.2

$\int_0^{\pi} x f(\sin x) dx$ equals

(A) $2\pi \int_0^{\frac{\pi}{2}} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

(B) $\pi \int_0^{\pi} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

(C) $\pi \int_0^{\frac{\pi}{2}} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

(D) None of these

Q.3

If $\int_0^{\pi} x f(\sin x) dx = A \int_0^{\pi/2} f(\sin x) dx$, then A is equal to:

(A) 0

(B) π

(C) $\frac{\pi}{4}$

(D) 2π

Q.4

If $f(a+b-x) = f(x)$, then $\int_a^b x f(x) dx$ is equal to

(A) $\frac{\mathrm{a}+\mathrm{b}}{2} \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{b}-\mathrm{x}) \mathrm{dx}$

(B) $\frac{\mathrm{a}+\mathrm{b}}{2} \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{b}+\mathrm{x}) \mathrm{dx}$

(C) $\frac{b-a}{2} \int_a^b f(x) dx$

(D) $\frac{\mathrm{a}+\mathrm{b}}{2} \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}$

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Q.5

The value of the integral $I = \int_0^{\frac{\pi}{4}} \sin x + \cos x \cos x - \sin x dx$ is equal to
(where, . denotes the greatest integer function)

(A) $\sqrt{2}$

(B) $2\sqrt{2}$

(C) 1

(D) $\sqrt{2} - 1$

Q.6

If $f(x) = \frac{e^x}{1+e^x}$, $I_1 = \int_{f(-a)}^{f(0)} x g\{x(1-x)\} dx$ and $I_2 = \int_{f(-a)}^{f(\theta)} g\{x(1-x)\} dx$, then the value of $\frac{I_2}{I_1}$ is

(A) 2

(B) 1

(C) -1

(D) -3

Q.7

The value of $\int_{-2}^{2} \left(ax^3 + bx + c\right) dx$ depends on

(A) b

(B) c

(C) a

(D) a and b

Q.8

What is $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{\sin^5 x \cos^3 x}{x^4} dx$ is equal to ?

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{8}$

(D) 0

Q.9

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$\int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \frac{|x|dx}{8\cos^2 2x+1}$ has the value

(A) $\frac{\pi^2}{6}$

(B) $\frac{\pi^2}{12}$

(C) $\frac{\pi^2}{24}$

(D) None of these

Q.10

If $f(x)$ is an even function, then what 1 s $\int_0^{\pi} f(\cos x)dx$ equal to?

(A) 0

(B) $\int_0^{\frac{\pi}{2}} f(\cos x)dx$

(C) $2\int_0^{\frac{\pi}{2}} f(\cos x)dx$

(D) 1

Q.11

The vatue of $\int_{-2}^{3} \left|1-\mathbf{x}^2\right| \mathrm{dx}$ is

(A) $\frac{28}{3}$

(B) $\frac{1}{3}$

(C) $\frac{7}{3}$

(D) $\frac{14}{3}$

Q.12

The value of $\int_0^{\pi} \frac{dx}{1+5^{\cos x}} + \int_{-2}^{2} \log\left(\frac{5-x}{5+x}\right) dx$ is

(A) $\pi/2$

(B) $\pi/4$

(C) π

(D) None of these

Q.13

The value of $\int_0^2 \mathbf{x}^{[x^2+1]}dx$, where $[x]$ is the greatest integer less than or equal to x is

(A) 2

(B) $8/3$

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(C) 4

(D) none of these

Q.14

The vahe of $\int_{-4}^{-5} e^{(x+5)^2} dx + 3\int_{1/3}^{2/3} e^{9\left(x-\frac{2}{3}\right)^2} dx$

(A) 0

(B) 1

(C) $1/2$

(D) $3/2$

Q.15

The value of $\int_{-1}^{3}\{[x] + |x-2|\}dx$

(A) 3

(B) 2

(C) 5

(D) 7

Q.16

The integral $\int_0^{\pi/2} |\sin x - \cos x| dx$ is equal to :

(A) $2\sqrt{2}$

(B) $2(\sqrt{2}-1)$

(C) $\sqrt{2}+1$

(D) None of these

Q.17

If $\int_{-3}^{2} f(x)dx = \frac{7}{3}$ and $\int_{-3}^{9} f(x)dx = -\frac{5}{6}$, then what is the value of $\int_{2}^{9} f(x)dx$?

(A) $-\frac{19}{6}$

(B) $\frac{19}{6}$

(C) $\frac{3}{2}$

(D) $-\frac{3}{2}$

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Q.18

$\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)}$ is

(A) $\frac{\pi ab}{a+b}$

(B) $\frac{\pi}{2(a+b)}$

(C) $\frac{\pi}{2ab(a+b)}$

(D) $\frac{\pi(a+b)}{2ab}$

Q.19

Let $f : \mathbf{R} \to \mathbf{R}$ and $g : \mathbf{R} \to \mathbf{R}$ be continuous functions. Then the value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \{f(x) + f(-x)\}\{g(x) - g(-x)\}dx$ is

(A) $f(x)g(x)$

(B) $f(x) + g(x)$

(C) 0

(D) none of these

Q.20

If m is an integer, then $\int_0^\pi \frac{\sin(2mx)}{\sin x}dx$ is equal to:

(A) 1

(B) 2

(C) 0

(D) π

Answers & Solutions

Q.1 Answer:

$\frac{3}{2}$

Solution:

$$\text{Let } I = \int_3^6 \frac{\sqrt{x}}{\sqrt{9-x}+\sqrt{x}}dx$$
$$I = \int_3^6 \frac{\sqrt{9-x}}{\sqrt{x}+\sqrt{9-x}}dx$$
$$2I = \int_3^6 (1)dx$$
$$I = \frac{3}{2}$$

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Q.2 Answer:

$\pi \int_0^{\frac{\pi}{2}} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

Solution:

Let $\mathrm{I}=\int_0^{\pi} \mathrm{xf}(\sin \mathrm{x}) \mathrm{dx} \ldots(1)$

Using property $\int_0^a f(x) dx=\int_0^a f(a-x) dx$

$\mathrm{I}=\int_0^{\pi}(\pi-\mathrm{x}) \mathrm{f}(\sin (\pi-\mathrm{x})) \mathrm{dx}=\int_0^{\pi}(\pi-\mathrm{x}) \mathrm{f}(\sin \mathrm{x}) \mathrm{dx} \ldots .$ **(2)**

Adding (1) and (2), we get

$2 \mathrm{I}=\pi \int_0^{\pi} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx} \Rightarrow \mathrm{I}=\frac{\pi}{2} \int_0^{\pi} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

Using property $\int_0^{2 a} \mathrm{f}(\mathrm{x}) \mathrm{dx}=\int_0^{\mathrm{a}} \mathrm{f}(\mathrm{x}) \mathrm{dx}+\int_0^{\mathrm{a}} \mathrm{f}(2 \mathrm{a}-\mathrm{x}) \mathrm{dx}$

$I=\frac{\pi}{2}\left(\int_0^{\pi / 2} f(\sin x) dx+\int_0^{\pi / 2} f(\sin (\pi-x)) dx\right)=\pi \int_0^{\pi / 2} f(\sin x) dx$

Q.3 Answer:

π

Solution:

Let $\mathrm{I}=\int_0^{\pi} \mathrm{xf}(\sin \mathrm{x}) \mathrm{dx} \ldots(1)$

$\mathrm{I}=\int_0^{\pi}(\pi-\mathrm{x}) \mathrm{f}(\sin (\pi-\mathrm{x})) \mathrm{dx}=\int_0^{\pi}(\pi-\mathrm{x}) \mathrm{f}(\sin \mathrm{x}) \mathrm{dx} \ldots(2)$

Adding (1) **and** (2)

$2 \mathrm{I}=\int_0^{\pi}(\mathrm{x}+\pi-\mathrm{x}) \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}=\pi \int_0^{\pi} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

$\Rightarrow 2 \mathrm{I}=2 \pi \int_0^{\pi / 2} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx} \Rightarrow \mathrm{I}=\pi \int_0^{\pi / 2} \mathrm{f}(\sin \mathrm{x}) \mathrm{dx}$

$\Rightarrow \mathrm{A}=\pi$

Q.4 Answer:

$\frac{\mathrm{a}+\mathrm{b}}{2} \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}$

Solution:

$\text{Let } \mathrm{I}=\int_{\mathrm{a}}^{\mathrm{b}} \mathrm{xf}(\mathrm{x}) \mathrm{dx} \ldots(i)$

$\therefore \mathrm{I}=\int_{\mathrm{a}}^{\mathrm{b}}\left(\mathrm{a}+\mathrm{b}-\mathrm{x} \cdot \ldots(\mathrm{x}) \mathrm{f}(\mathrm{a}+\mathrm{b}-\mathrm{x}) \mathrm{dx}, \left(\because \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}=\int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{a}+\mathrm{b}-\mathrm{x}) \mathrm{dx}\right)\right.$

$\Rightarrow \mathrm{I}=\int_{\mathrm{a}}^{\mathrm{b}}(\mathrm{a}+\mathrm{b}-\mathrm{x}) \mathrm{f}(\mathrm{x}) \mathrm{dx}$

$\Rightarrow \mathrm{I}=(\mathrm{a}+\mathrm{b}) \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}-\mathrm{I}[\text{Using}(1)]$

$\Rightarrow \mathrm{I}+\mathrm{I}=(\mathrm{a}+\mathrm{b}) \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}$

$\Rightarrow 2 \mathrm{I}=(\mathrm{a}+\mathrm{b}) \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}$

$\Rightarrow \mathrm{I}=\left(\frac{\mathrm{a}+\mathrm{b}}{2}\right) \int_{\mathrm{a}}^{\mathrm{b}} \mathrm{f}(\mathrm{x}) \mathrm{dx}$

Hence, the correct Answer is D.

Q.5 Answer:

$\sqrt{2}$ - 1

Solution:

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$I = \int_0^{\frac{\pi}{4}} \sin x + \cos x \cos x - \sin x dx$

Let, $\sin x + \cos x = t$

$\Rightarrow I = \int_1^{\sqrt{2}} t dt = \int_1^{\sqrt{2}} 1 dt = \sqrt{2} - 1$

Q.6 Answer:

2

Solution:

$f(a) + f(-a) = 1$

$= \int_{f(-a)}^{f(a)} (f(a) + f(-a) - x) g((1-x)x\} dx$

$= \int_{f(-a)}^{f(a)} (1-x) g\{(1-x)x\} dx$

$= I_2 - \int_{f(-a)}^{f(a)} x g\{x(1-x)\} dx$

$2I_1 = I_2$

Therefore, $\frac{I_2}{I_1} = 2$

Q.7 Answer:

c

Solution:

$I = \int_{-2}^{2} \left(ax^3 + bx\right) dx + c \quad \int_{-2}^{2} \left(\left(ax^3 + bx\right) + c\right) dx = 0 + c.4 = 4c$

$\left(\because ax^3 + bx \text{ is an odd function of } x\right)$

Q.8 Answer:

0

Solution:

Let $f(x) = \frac{\sin^5 x \cos^3 x}{x^4}$

$f(-x) = \frac{\sin^5(-x)\cos^3(-x)}{(-x)^4}$

$= \frac{-\sin^5 x \cos^3 x}{x^4} = -f(x)$

$\Rightarrow$ f(x) **is an odd function.**

Hence, $\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{\sin^5 x \cos^3 x}{x^4} dx = 0$

Q.9 Answer:

$\frac{\pi^2}{12}$

Solution:

Let f(x) $= \frac{|x|}{8\cos^2 2x+1}$

then $f(-x) = \frac{|-x|}{8\cos^2 2(-x)+1} = \frac{|x|}{8\cos^2 2x+1}$

= f(x)

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$f(x)$ **is even function**

$\therefore I = \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \frac{|x|dx}{8\cos^2 2x+1} = 2\int_0^{\frac{\pi}{2}} \frac{|x|dx}{8\cos^2 2x+1}$

$= 2\int_0^{\frac{\pi}{2}} \frac{xdx}{8\cos^2 2x+1} = 2I_1$

Now $\therefore I_1 = \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2}-x\right)dx}{8\cos^2 2\left(\frac{\pi}{2}-x\right)+1} = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}-x}{8\cos^2 2x+1}dx$

$= \frac{\pi}{2}\int_0^{\frac{\pi}{2}} \frac{\mathrm{dx}}{8\cos^2 2\mathrm{x}+1} - \mathrm{I}_1$

$2\mathrm{I}_1 = \frac{\pi}{2}\cdot 2\int_0^{\frac{\pi}{4}} \frac{\mathrm{dx}}{8\cos^2 2\mathrm{x}+1}$

$= \pi\int_0^{\frac{\pi}{4}} \frac{\sec^2 2x}{9+\tan^2 2x}dx$

$\mathrm{Put}\ \tan 2x = t \Rightarrow 2\sec^2 2xdx = dt$

$2\mathrm{I}_1 = \frac{\pi}{2}\int_0^{\infty} \frac{\mathrm{dt}}{9+\mathrm{t}^2} = \frac{\pi}{2}\cdot\frac{1}{3}\left[\tan^{-1}\frac{\mathrm{t}}{3}\right]_0^{\infty} = \frac{\pi}{2}\cdot\frac{1}{3}\cdot\frac{\pi}{2}$

$\therefore \mathrm{I}_1 = \frac{\pi^2}{24} \Rightarrow \mathrm{I} = 2\mathrm{I}_1 = \frac{\pi^2}{12}$

Q.10 Answer:

$2\int_0^{\frac{\pi}{2}} f(\cos x)dx$

Solution:

Since $f(x)$ **is an even function therefore** $\int_0^{\pi} f(x)dx = 2\int_0^{\pi/2} f(x)dx$

Hence, $\int_0^{\pi} f(\cos x)dx = 2\int_0^{\pi/2} f(\cos x)dx$

Q.11 Answer:

$\frac{28}{3}$

Solution:

$\int_{-2}^{3} \left|1-x^2\right| dx$

$\int_{-2}^{-1} \left(x^2-1\right) dx + \int_{-1}^{1} \left(1-x^2\right) dx + \int_{1}^{3} \left(x^2-1\right) dx = \frac{28}{3}$

Q.12 Answer:

$\pi/2$

Solution:

Let $I = I_1 + I_2$

Consider $I_1 = \int_0^{\pi} \frac{dx}{1+5^{\mathrm{cois}\,x}}$

Now $I_1 = \int_0^{5} \frac{dx}{1+5^{\cos(x-x)}} = \int_0^{\pi} \frac{dx}{1+5^{-\infty\times xx}} = \int_0^{\pi} \frac{5^{\cos xx}dx}{5^{\mathrm{cosex}}+1} \ldots (2)$

Adding (1) and (2), we get

$2I_1 = \int_0^{\pi} \frac{dx}{1+5^{\infty\times 8x}} + \int_0^{\pi} \frac{5^{\mathrm{con}\,x}dx}{5^{\cos x}+1} = \int_0^{\pi} 1.dx = \pi$

$I_1 = \pi/2$

Consider $I_2 = \int_{-2}^{2} \log\left(\frac{5-x}{5+x}\right) dx$

Let $g(x) = \log\left(\frac{5-x}{5+x}\right)$

Now $g(-x) = \log\left(\frac{5-(-x)}{5+(-x)}\right) = -\log\frac{5-x}{5+x} = -g(x)$

$\therefore g(x)$ **is an odd function**

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$\therefore \int_{-2}^{3} g(x)dx = 0 \Rightarrow \mathrm{I}_2 = 0$
$\mathrm{I} = \mathrm{I}_1 + \mathrm{I}_2 = \pi/2 + 0 = \pi/2$

Q.13 Answer:

none of these

Solution:

For $x \in [0,2], x^2+1 \in [1,5]$
we must break $[0,2]$
$= [0,1] \cup [1,\sqrt{2}] \cup [\sqrt{2},\sqrt{3}] \cup [\sqrt{3},2]$
Hence $\int_0^2 x^{\left[x^2+1\right]}dx$
$= \int_0^1 x^{\left[x^2+1\right]}dx + \int_1^{\sqrt{2}} x^{\left[x^2+1\right]}dx +$
$\int_{\sqrt{2}}^{\sqrt{3}} x^{\left[x^2+1\right]}dx + \int_{\sqrt{3}}^2 x^{\left[x^2+1\right]}dx$
$= \int_0^1 xdx + \int_0^{\sqrt{2}} dx + \int_{\sqrt{3}}^{\sqrt{3}} x^3dx + \int_{\sqrt{3}}^2 x^4dx$
$= \frac{1}{2} + \frac{1}{3}\left[2^{3/2}-1\right] + \frac{1}{4}[9-4] + \frac{1}{5}\left[32-3^{5/2}\right] = \frac{469}{60} + \frac{1}{3}2^{3/2} - \frac{1}{5}3^{5/2}$

Q.14 Answer:

0

Solution:

$I_1 = \int_4^{-5} e^{(x+5)^2}dx = (5-4)\int_0^1 e^{((-s+4)z-4+s)^2}dx$
$I_1 = \int_0^1 e^{(x-1)^2}dx$
Again let $I_2 = \int_{1/3}^{2/3} e^{9\left(x-\frac{2}{3}\right)^2}dx$
$I_2 = \left(\frac{2}{3}-\frac{1}{3}\right)\int_0^1 e^{\delta\left[\left(\frac{2}{3}\frac{1}{2}\right)^{x+\frac{1}{3}}\frac{2}{3}\right]^2}dx$
$= \frac{1}{3}\int_0^1 \theta^{(x-1)^2}dx = \frac{1}{3}(-l) \quad \ldots\ldots\ldots (ii)$
where $\mathbf{I} = \mathrm{I}_1 + 3\mathrm{I}_2$
$= \mathrm{I}_1 + 3\left(-\frac{1}{3}\right) = \mathrm{I}_1 - \mathrm{I}_1$
$\mathbf{I = 0}$
$\int_4^{-5} e^{(x+6)^2}dx + 3\int_{13}^{2/3} e^{\left(x\frac{2}{3}\right)^2}dx = 0$

Q.15 Answer:

7

Solution:

$\int_{-3}^{3}\{[x] + |x-2|\}dx$
$\int_{-1}^{3}[x]dx + \int_{-1}^{3}|x-2|dx$
$= \int_{-1}^{0}[x]dx + \int_0^1[x]dx + \int_1^2[x]dx + \int_2^3[x]dx$
$+ \int_{-1}^{2}|x-2|dx + \int_2^3|x-2|dx$
$= -1 + 0 + 1 + 2 + \frac{9}{2} + \frac{1}{2} = 7$

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Q.16 Answer:

$2(\sqrt{2}-1)$

Solution:

$\int_0^{\pi/4} \cos x - \sin x$
$+ \int_{\pi/4}^{\pi/2} \sin x - \cos x$
$= 2(\sqrt{2}-1)$

Q.17 Answer:

$-\frac{19}{6}$

Solution:

Value of the integral $\int_2^9 \mathrm{f(x)dx}$
$= \int_{-3}^9 \mathrm{f(x)dx} - \int_{-3}^2 \mathrm{f(x)dx}$
Given, $\int_{-3}^9 \mathrm{f(x)dx} = \frac{-5}{6}$ **and** $\int_{-3}^2 \mathrm{f(x)dx} = \frac{7}{3}$
Putting these values in equation (i) $\int_2^9 f(x)dx = \frac{-5}{6} - \frac{7}{3} = -\frac{19}{6}$

Q.18 Answer:

$\frac{\pi}{2ab(a+b)}$

Solution:

$\int_0^\infty \frac{dx}{(x^2+a^2)(x^2+b^2)}$
$= \frac{1}{b^2-a^2} \int_0^\infty \frac{\left(x^2+b^2\right)-\left(x^2+a^2\right)}{(x^2+a^2)(x^2+b^2)}$
$= \frac{1}{b^2-a^2} \int_0^\infty \left[\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2}\right] dx$
$= \frac{1}{b^2-a^2} \left[\frac{1}{a}\tan^{-1}\frac{x}{a} - \frac{1}{b}\tan^{-1}\frac{x}{b}\right]_0^\infty$
$= \frac{1}{b^2-a^2} \left[\frac{\pi}{2a} - \frac{\pi}{2b}\right] = \frac{\pi}{2ab(a+b)}$

Q.19 Answer:

0

Solution:

Let $\phi(\mathrm{x}) = \{\mathrm{f(x)} + \mathrm{f(-x)}\}\{\mathrm{g(x)} - \mathrm{g(-x)}\}$
$\phi(-x) = \{f(-x) + f(x)\}\{g(-x) - g(x)\}$
$= -\{\mathrm{f(x)} + \mathrm{f(-x)}\}\{\mathrm{g(x)} - \mathrm{g(-x)}\} = -\phi(\mathrm{x})$
$\therefore \varphi(\mathrm{x})$ **is an odd function** $\Rightarrow \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \phi(\mathrm{x})\mathrm{dx} = 0$

Q.20 Answer:

0

Solution:

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Use $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

$\int_0^\pi \frac{\sin 2mx}{\sin x}dx = \int_0^\pi \frac{\sin(2m\pi - 2mx)}{\sin(\pi - x)}dx$

$= \int_0^\pi \frac{-\sin 2mx}{\sin x}\mathrm{dx} = -\mathrm{I} \Rightarrow 2\mathrm{I} = 0 \Rightarrow \mathrm{I} = 0$

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